“CIVIC COMPLAINT CLASSIFICATION SYSTEM USING DISTILBERT”

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Introduction

Traditional civic complaint systems are often manual, slow, and inefficient, leading to delays and incorrect complaint handling. Citizens face a lot of challenges in communication issues to the appropriate authorities. This system is an AI-based system that uses NLP and DistilBERT to automate complaint classification and improve service efficiency.

Objective

To develop a mobile application that allows users to submit civic complaints via text and classify and route them to the appropriate authority using Distilbert.

Methodology

The system works by allowing users to submit civic complaints through a mobile application using text in English or Nepali along with their GPS location. The submitted text is first preprocessed using basic NLP techniques such as cleaning and tokenization to make it suitable for analysis. It is then passed to a fine-tuned DistilBERT model, a transformer-based NLP model that understands the context and meaning of the text, which automatically classifies the complaint into one of the predefined categories: electricity, water, road, or garbage.
The model is trained on labeled complaint data and uses deep learning techniques to accurately identify patterns in user input. It can handle both structured and unstructured text, making it effective even for real-world complaints that may be short or ambiguous.
After classification, the backend system developed using Django stores the complaint and routes it to the appropriate authority based on its category and location. The system also provides real-time notifications and status updates to authorities, ensuring efficient tracking and faster resolution of complaints.

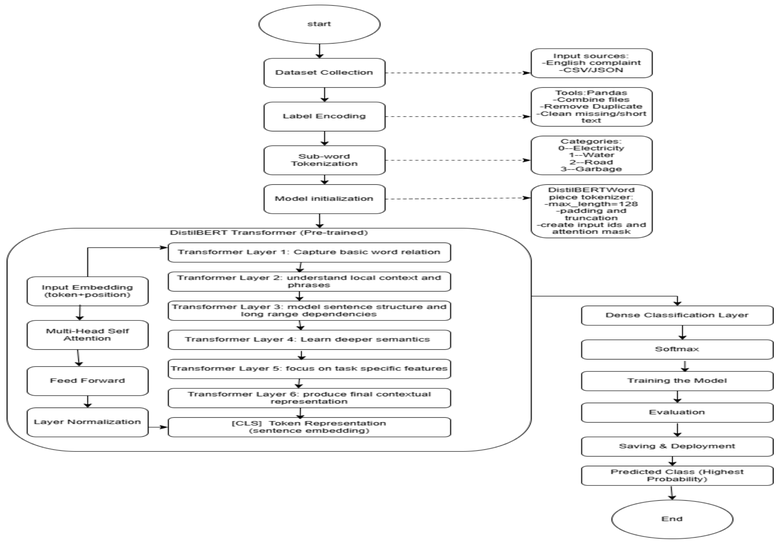

Figure 1. High-level architecture of the AI-Based Complaint Classification Module illustrating the end-to-end pipeline from raw complaint data to category prediction

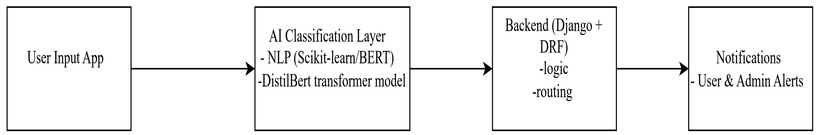

Figure 2. Block Diagram

Results and Analysis

The system performs well in classifying civic complaints for both English and Nepali. Confusion matrices for regular and hard tests show high accuracy (≈95–100%), with most predictions correctly classified. Accuracy is slightly lower on hard test data (F1 slightly < 1.0), especially for short or ambiguous complaints. Comparison of English and Nepali models on the hard test dataset shows similar performance with minor differences, with the English model achieving scores close to 1.0 and the Nepali model slightly lower. Overall, the model is reliable and effective for real-world deployment.

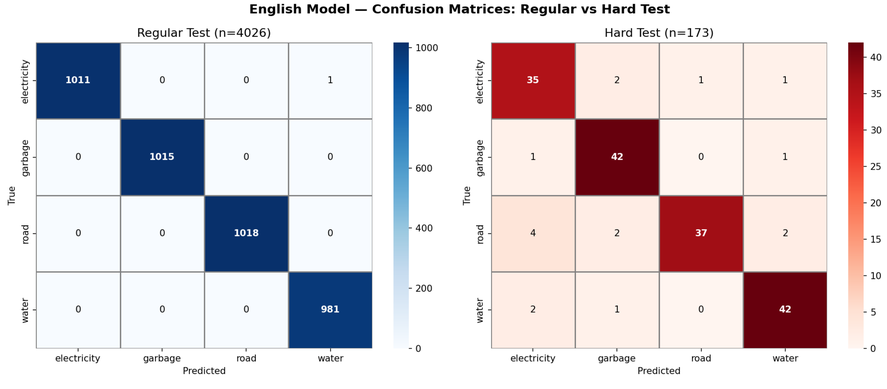

Figure 3. Confusion Matrix for English Model

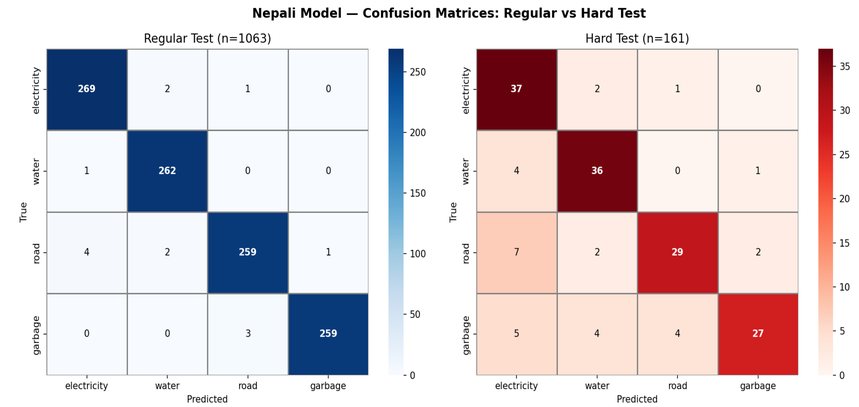

Figure 4. Confusion Matrix for Nepali Model

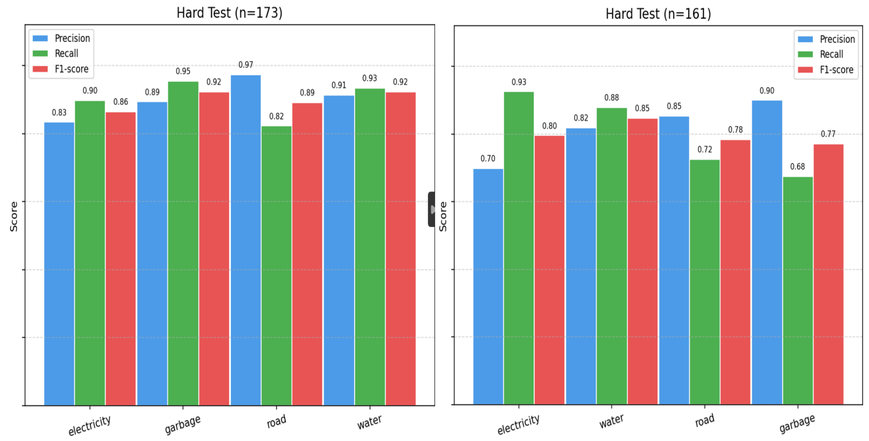

Figure 5. Hard Test Class Metrics for English VS Nepali model

Conclusion

The system successfully classifies civic complaints in both English and Nepali using a fine-tuned DistilBERT model. It accurately routes complaints to the appropriate authorities and provides real-time notifications to users. Performance remains strong even on challenging inputs, demonstrating that the system is reliable, multilingual, and suitable for real-world deployment.

References

- **Welch, T. A. (1984). *A Technique for High-Performance Data Compression.* IEEE Computer.**
- **Maharjan, S., Poudel, S., & Tandukar, D. (2024). *A Comparative Study of Text-Based Lossless Compression.* American Journal of Smart Technology and Solutions.**